

FINAL REPORT – GYM MANAGEMENT SYSTEM

PRESENTATION LINK :

https://www.canva.com/design/DAGmCVFIV0U/xM8fop554b7WqNkzyz0E2g/edit?utm_content=DAGmCVFIV0U&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

1. User Stories

Member 1: [Fahad Ahmad]

1. As a **new user**, I want to **sign up easily** so that I can access workout and diet plans.
 2. As a **registered user**, I want to **log in securely** to access my personalized dashboard.
 3. As a **user**, I want to **view available workout plans** so I can choose one suitable for my fitness goal.
 4. As a **user**, I want to **book a session with a trainer** to get personalized training advice.
 5. As a **user**, I want to **view my diet plans** to follow the nutrition guidelines provided.
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Member 2: [Taha Faisal]

6. As an **admin**, I want to **approve new user registrations** so only valid users can access the system.
 7. As an **admin**, I want to **add and manage trainers** to provide users with more training options.
 8. As an **admin**, I want to **monitor user membership renewals** to maintain the gym's active member list.
 9. As an **admin**, I want to **manage available workout and diet plans** so users have updated options.
 10. As a **user**, I want to **view available trainers' schedules** to choose the best session timing.
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Member 3: [Sohaib Imran]

11. As a **user**, I want to **renew my gym membership online** to avoid visiting the gym physically.
 12. As a **user**, I want to **see my payment history** to track my spending.
 13. As a **trainer**, I want to **view my assigned training sessions** to manage my availability.
 14. As an **admin**, I want to **view pending user requests** to approve or reject membership.
 15. As a **user**, I want to **receive a confirmation notification** after booking a trainer session.
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2. Functional and Non-Functional Requirements

Functional Requirements:

- Users can register, login, and logout securely.
 - Admins can approve/reject user registrations.
 - Admins can add/edit/delete trainers.
 - Users can view workout and diet plans.
 - Users can book trainers for sessions.
 - Payments can be made for membership renewals.
 - Admins can monitor trainer schedules.
 - Users can view their dashboards and payment histories.
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Non-Functional Requirements:

- The system must respond within **3 seconds** for any user action.
 - The system must support at least **100 concurrent users**.
 - Passwords must be **encrypted** during storage.
 - System should be **available 99%** of the time.
 - Data must be backed up **daily** to prevent loss.
 - System must be **scalable** to accommodate future enhancements.
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3. Project Plan

Phase	Activities	Timeline (April 20–April 29)
Requirements Gathering	User stories, system requirements	April 20–21
System Design	ERD, Wireframes, Architecture	April 22
Development	Frontend (HTML Templates), Backend (Spring Boot Controllers, Services)	April 22–26
Testing	Black Box, White Box Testing, Coverage Reports	April 27
Deployment	Local Server Deployment and Testing	April 28
Documentation	Final reports and submission	April 29

Simple Architecture Flow:

- User interacts with the **frontend**.
 - Requests are processed by **controllers**.
 - Business logic is handled by **services**.
 - Data access through **repositories**.
 - Data stored/retrieved from **database**.
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4. Design

System Design Overview:

- **Frontend:** HTML Templates
- **Backend:** Spring Boot Controllers, Services, Repositories
- **Database:** MySQL/PostgreSQL
- **Security:** Basic authentication and password encryption

Main Pages:

- User Login / Signup
 - Admin Login / Signup
 - User Dashboard
 - Admin Dashboard
 - Trainer Management
 - Workout Plan Viewing
 - Diet Plan Viewing
 - Payment Page
 - Trainer Booking Page
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5. Work Division between 3 Members

Member

Work Assigned

Member 1 User Authentication Module, User Dashboard, Diet Plan Module

Member 2 Admin Module (Admin Dashboard, Trainer Management), Workout Plan Module

Member 3 Payment System, Trainer Booking System, Testing and Documentation

UML Diagrams

Architecture Diagram - Gym Management System

